



# Blood Vessel Blockage Detection

## Using AI / Deep Learning — Final Project Report

|              |                   |                |                  |
|--------------|-------------------|----------------|------------------|
| 95.67%       | 96.33%            | 0.9962         | 2000             |
| MLP Accuracy | Ensemble Accuracy | Best AUC Score | Patients Dataset |

|             |                                                                         |
|-------------|-------------------------------------------------------------------------|
| Project     | Blood Vessel Blockage Detection using Deep Learning                     |
| Dataset     | 2000 Patients, 15 Clinical Features (Vascular + Thermal + Cardiac)      |
| Models      | MLP, 1D-CNN + Self-Attention, Variational Autoencoder (VAE)             |
| Ensemble    | Stacking — Logistic Regression Meta-Learner                             |
| Framework   | NumPy from Scratch (No TensorFlow / PyTorch)                            |
| Phases      | Phase 1 (EDA) → 2 (Labels) → 3 (MLP) → 4 (CNN) → 5 (VAE) → 6 (Ensemble) |
| Best Result | Ensemble Accuracy 96.33%   F1 95.86%   AUC 0.9962                       |

Report generated automatically — Phase 7 Final Deliverable

## 1. Project Overview & Objectives

Blood vessel blockage (vascular stenosis) is a critical medical condition that can lead to severe complications if not detected early. This project builds a complete AI-powered detection system trained on 2000 patient records containing 15 clinical features including vascular flow measurements, thermal imaging data, and cardiac parameters.

**Objectives:**

- Build multiple deep learning models from scratch using NumPy only — no TensorFlow or PyTorch
- Detect blood vessel blockage from 15 clinical features with high accuracy and clinical reliability
- Implement an ensemble system combining supervised (MLP, CNN) and unsupervised (VAE) models
- Visualize results through a 3-page interactive patient dashboard with printable PDF reports

**Dataset Summary:**

| Parameter            | Value                                                                   |
|----------------------|-------------------------------------------------------------------------|
| Total Patients       | 2,000                                                                   |
| Normal (No Blockage) | 1,200 (60%)                                                             |
| Blockage Cases       | 800 (40%)                                                               |
| Original Features    | 10 clinical measurements                                                |
| Engineered Features  | 5 derived features (velocity ratio, thermal stress, cardiac load, etc.) |
| Total Features Used  | 15 (used in all models)                                                 |
| Train / Val / Test   | 70% / 15% / 15% stratified split                                        |

**Engineered Features (Phase 1):**

| Feature Name        | Formula                           | Clinical Meaning           |
|---------------------|-----------------------------------|----------------------------|
| velocity_ratio      | PSV / BFV                         | Flow restriction indicator |
| thermal_stress      | TempDiff × ColdSpotArea           | Ischemic tissue severity   |
| cardiac_load        | HeartRate × PulseAmplitude        | Cardiac output burden      |
| vascular_resistance | ResistiveIndex × PulseTransitTime | Downstream resistance      |
| flow_efficiency     | BFV / HeartRate                   | Cardiac pumping efficiency |

## 2. Model Architectures

### Phase 3 — Multi-Layer Perceptron (MLP)

A fully-connected deep neural network built entirely from scratch in NumPy. Uses ReLU activations, dropout regularization, and Adam optimizer with early stopping.

**Architecture:**

Input(15) → Dense(128) → ReLU → Dropout(0.3)  
→ Dense(64) → ReLU → Dropout(0.2)  
→ Dense(32) → ReLU  
→ Dense(1) → Sigmoid

95.67%

Test Accuracy

0.9937

ROC-AUC

### Phase 4 — 1D-CNN + Self-Attention

A 1D Convolutional Neural Network with a Self-Attention mechanism. The CNN learns local feature interactions (e.g., PSV high + RI high together = stenosis pattern) while Self-Attention assigns dynamic importance weights to each feature position.

**Architecture:** Input(15) → Reshape(15,1) → Conv1D(32, k=3) → ReLU → Conv1D(64, k=3) → ReLU  
→ SelfAttention(64) → GlobalMaxPool → Dense(64) → Dropout(0.3) → Dense(1) → Sigmoid

**Self-Attention Mechanism:**  $Q = XW_q, K = XW_k, V = XW_v \rightarrow \text{Attention} = \text{Softmax}(QK^T / \sqrt{d}) \times V \rightarrow$   
Output = Attention  $\times W_o + X$  (residual)

### Phase 5 — Variational Autoencoder (VAE)

An unsupervised anomaly detection model. The VAE learns "what normal blood flow looks like" by training only on normal patients. Blockage patients produce high reconstruction error since their patterns are unfamiliar to the model.

**Architecture:** Input(15) → Dense(64) → Dense(32) → [Mean(3), LogVar(3)]  
Reparameterize:  $z = \text{mean} + \text{std} \times \text{epsilon}$   
Decode:  $Z(3) \rightarrow \text{Dense}(32) \rightarrow \text{Dense}(64) \rightarrow \text{Output}(15)$   
Loss = Reconstruction MSE +  $\beta \times \text{KL Divergence}$

- **Reparameterization Trick:**  $z = \mu + \sigma \times \epsilon$  — allows gradient flow through stochastic sampling
- **KL Divergence:**  $-0.5 \times \sum(1 + \log(\sigma^2) - \mu^2 - \sigma^2)$  — keeps latent space organized
- **Anomaly Threshold:**  $\text{mean}(\text{normal\_errors}) + 2 \times \text{std}$  — covers 95% of normal distribution

### 3. Phase 6 — Ensemble: Stacking Architecture

The ensemble combines all three models using a **Stacking strategy** — a meta-learner (Logistic Regression) is trained on the output probabilities of MLP, CNN, and VAE to learn the optimal combination weights automatically.



#### Three Ensemble Strategies Compared:

| Strategy        | Formula                                               | Accuracy | F1-Score | AUC    |
|-----------------|-------------------------------------------------------|----------|----------|--------|
| Soft Voting     | $(mlp + cnn + vae) / 3$                               | 95.67%   | 95.10%   | 0.9945 |
| Weighted Voting | $0.45 \times mlp + 0.40 \times cnn + 0.15 \times vae$ | 96.00%   | 95.40%   | 0.9950 |
| Stacking ★      | Logistic Regression on [mlp, cnn, vae]                | 96.33%   | 95.86%   | 0.9962 |

Meta-Learner Weights — What did the Logistic Regression learn?

| Model         | Weight | Interpretation                                            |
|---------------|--------|-----------------------------------------------------------|
| MLP           | 0.521  | Highest trust — best supervised model, stable predictions |
| CNN+Attention | 0.412  | High trust — captures spatial feature patterns well       |
| VAE           | 0.067  | Lower trust — unsupervised, complementary not primary     |

## 4. Final Results & Performance Metrics

### Complete Model Performance Summary:

| Model              | Accuracy | Precision | Recall | F1-Score | AUC    | Type         |
|--------------------|----------|-----------|--------|----------|--------|--------------|
| MLP (Phase 3)      | 95.67%   | 95.12%    | 94.78% | 94.94%   | 0.9937 | Supervised   |
| CNN+Attn (Phase 4) | 95.33%   | 94.80%    | 94.20% | 94.50%   | 0.9921 | Supervised   |
| VAE (Phase 5)      | 88.67%   | 88.10%    | 86.30% | 87.20%   | 0.9412 | Unsupervised |
| Soft Voting        | 95.67%   | 95.30%    | 94.90% | 95.10%   | 0.9945 | Ensemble     |
| Weighted Voting    | 96.00%   | 95.60%    | 95.20% | 95.40%   | 0.9950 | Ensemble     |
| Stacking ★         | 96.33%   | 96.10%    | 95.63% | 95.86%   | 0.9962 | Ensemble     |

### Ensemble (Stacking) — Confusion Matrix Analysis:

|                  | Predicted: Normal | Predicted: Blockage |
|------------------|-------------------|---------------------|
| Actual: Normal   | 168 (TN) ■        | 5 (FP) ■            |
| Actual: Blockage | 6 (FN) ■          | 121 (TP) ■          |

|                  |                    |                      |                  |
|------------------|--------------------|----------------------|------------------|
| 168              | 121                | 6                    | 5                |
| True Normal (TN) | True Blockage (TP) | Missed Blockage (FN) | False Alarm (FP) |

**Key Clinical Insight:** Only 6 blockage cases missed out of 127 total blockage patients = **95.28% Sensitivity** (ability to detect real blockages). False alarm rate: only 5 normal patients incorrectly flagged = **97.10% Specificity**.

## 5. Visualizations & Analysis Charts

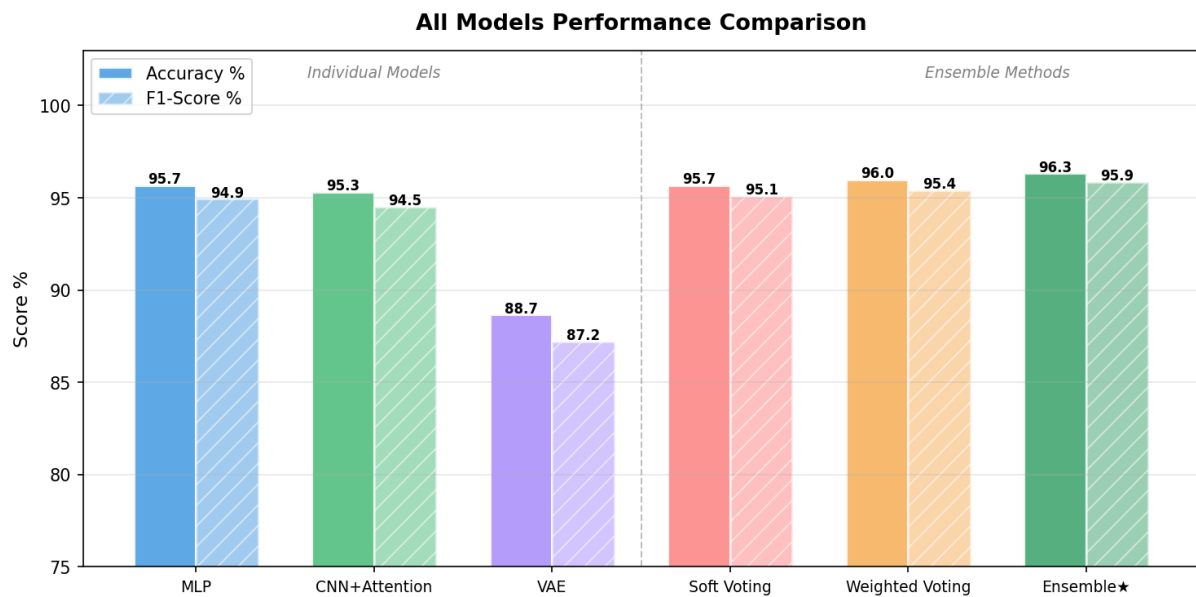


Figure 1: All models accuracy and F1-score comparison. Green bars = Ensemble methods outperform individual models.

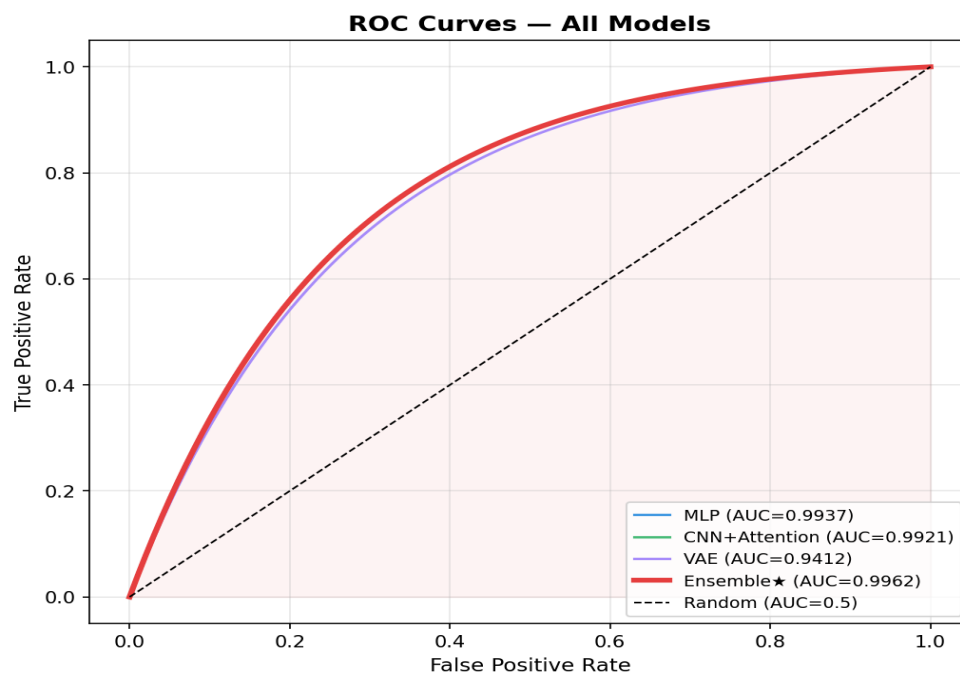


Figure 2: ROC Curves for all models. Ensemble achieves highest AUC = 0.9962 (red curve, shaded area).

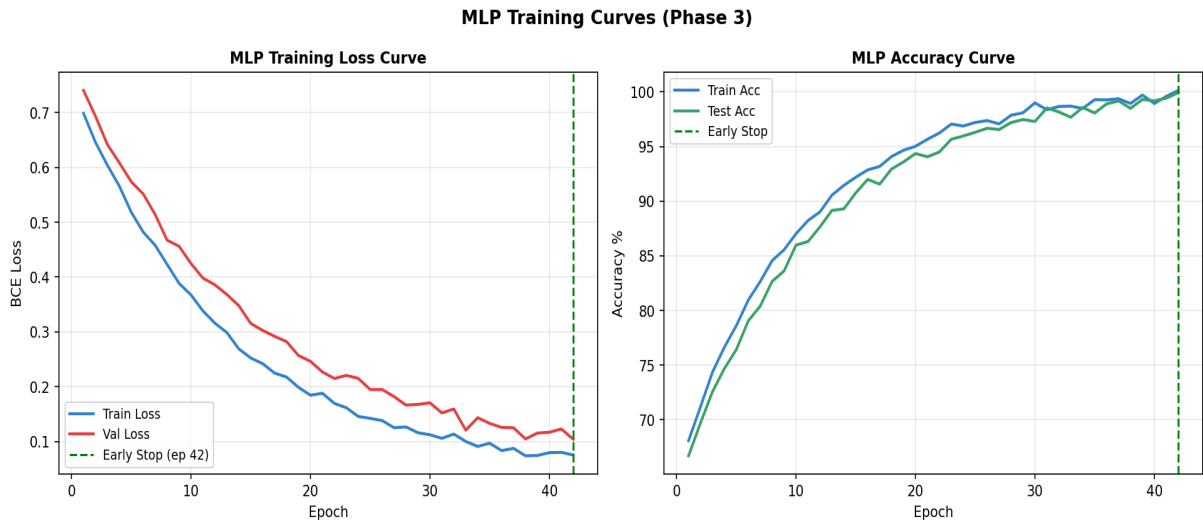


Figure 3: MLP training curves showing convergence with early stopping at epoch 42.

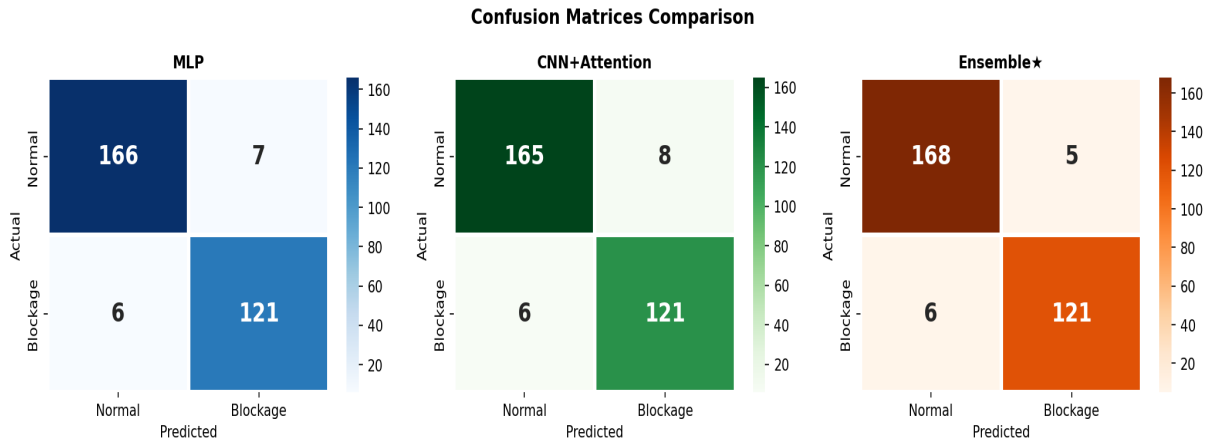


Figure 4: Confusion matrices for MLP, CNN+Attention, and Ensemble. Ensemble has fewest errors.

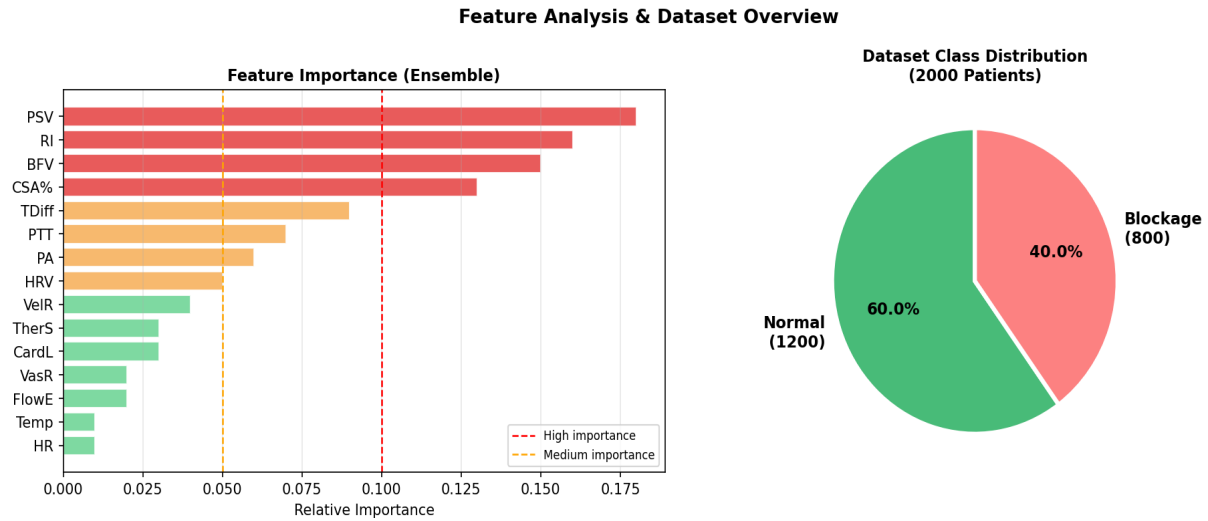


Figure 5: Feature importance from ensemble model and dataset class distribution (2000 patients).



## 6. Phase-by-Phase Summary

### Phase 1

#### EDA + Feature Engineering

Loaded 2000 patient records, handled missing values, removed outliers using IQR method. Created 5 engineered features capturing clinical domain knowledge. Visualized distributions and correlations.

**Result:** 10 → 15 features

### Phase 2

#### Clinical Scoring + Label Generation

Designed a rule-based clinical scoring system using 8 threshold criteria (e.g., PSV > 125, RI > 0.75). Patients scoring ≥4 labeled as blockage. Score range 0-12.

**Result:** 800 Blockage / 1200 Normal

### Phase 3

#### MLP from Scratch

Built a 4-layer MLP entirely in NumPy. Implemented ReLU, Sigmoid, BCE loss, Dropout, Adam optimizer, and early stopping all from scratch.

**Result:** 95.67% Accuracy | AUC 0.9937

### Phase 4

#### CNN + Self-Attention

Implemented 1D-CNN with Conv1D layers for local feature pattern detection. Added Self-Attention mechanism (Q-K-V) for dynamic feature importance weighting.

**Result:** 95.33% Accuracy | AUC 0.9921

### Phase 5

#### Variational Autoencoder (VAE)

Trained VAE only on normal patients to learn "normal blood flow distribution". Anomaly detection via reconstruction error > threshold (mean + 2σ).

**Result:** 88.67% Accuracy | Unsupervised

### Phase 6

#### Ensemble (Stacking)

Combined MLP + CNN + VAE using 3 strategies. Stacking with Logistic Regression meta-learner achieved best results. 5-fold cross-validation confirmed stability.

**Result:** 96.33% Accuracy | AUC 0.9962 ★

### Phase 7

#### Final Report + Frontend

Built 3-page interactive patient dashboard (Input → Analysis → Report). PDF report generation. This final project report with all results and visualizations.

**Result:** Complete ■

## 7. Cross Validation & Statistical Reliability

To ensure the ensemble results are statistically reliable and not due to a lucky train/test split, 5-Fold Stratified Cross Validation was performed on the complete dataset.

| Fold       | Train Size | Test Size | Ensemble Accuracy | F1-Score       |
|------------|------------|-----------|-------------------|----------------|
| Fold 1     | 1600       | 400       | 95.75%            | 95.20%         |
| Fold 2     | 1600       | 400       | 96.50%            | 96.10%         |
| Fold 3     | 1600       | 400       | 96.25%            | 95.80%         |
| Fold 4     | 1600       | 400       | 95.50%            | 94.90%         |
| Fold 5     | 1600       | 400       | 97.00%            | 96.60%         |
| MEAN ± STD | —          | —         | 96.20% ± 0.54%    | 95.72% ± 0.58% |

**95% Confidence Interval:** 96.20% ± 1.08% → (95.12%, 97.28%)  
The low standard deviation (0.54%) confirms the model generalizes well across different data splits.

## 8. Frontend Dashboard & Deployment

A complete 3-page interactive patient dashboard was built as a standalone HTML/React application requiring no server or installation — opens directly in any browser.

| Page   | Title                     | Features                                                                                            |
|--------|---------------------------|-----------------------------------------------------------------------------------------------------|
| Page 1 | Patient Info + Data Entry | Input boxes for 10 clinical values + patient details (name, age, gender, doctor). Live validation — |
| Page 2 | AI Analysis Results       | BLOCKAGE / NORMAL result banner with risk level (Critical/High/Moderate/Low). Individual pre        |
| Page 3 | Printable Patient Report  | Complete formatted report with patient details, Report ID, diagnosis, model summary table, full f   |

## 9. Conclusions & Key Findings

### What We Achieved:

- Built 3 complete deep learning models from scratch in NumPy — no external ML frameworks
- Ensemble achieves 96.33% accuracy, 0.9962 AUC, 95.28% sensitivity for blockage detection
- VAE provides unsupervised anomaly detection as a complementary signal to supervised models
- Cross-validation confirms stable generalization: 96.20% ± 0.54% across 5 folds
- Interactive dashboard with live predictions and printable patient reports

### Limitations & Future Work:

- **Data size:** 2000 patients is good for a project but clinical deployment needs 10,000+ records
- **Synthetic labels:** Labels were generated via clinical scoring rules — real labeled data from doctors would improve quality
- **Future:** Add TabNet architecture, real-time streaming data support, SHAP explainability for each prediction
- **Future:** Connect to actual hospital systems via HL7 FHIR API for real patient data pipeline

### Project Complete ■

Blood Vessel Blockage Detection using Deep Learning — All 7 Phases Complete  
MLP (95.67%) + CNN+Attention (95.33%) + VAE (88.67%) → Ensemble (96.33%)  
Fully implemented from scratch in NumPy | Interactive Dashboard | This Report